Dormand–Prince RK5(4) (used by `ode45`) for $y' = f(t, y)$, step h:

$$
\begin{aligned}
k_1 &= f\big(t_n,\ y_n\big) \\
k_2 &= f\big(t_n + \tfrac{1}{5}h,\ y_n + h\,\tfrac{1}{5}k_1\big) \\
k_3 &= f\big(t_n + \tfrac{3}{10}h,\ y_n + h\big(\tfrac{3}{40}k_1 + \tfrac{9}{40}k_2\big)\big) \\
k_4 &= f\big(t_n + \tfrac{4}{5}h,\ y_n + h\big(\tfrac{44}{45}k_1 - \tfrac{56}{15}k_2 + \tfrac{32}{9}k_3\big)\big) \\
k_5 &= f\big(t_n + \tfrac{8}{9}h,\ y_n + h\big(\tfrac{19372}{6561}k_1 - \tfrac{25360}{2187}k_2 + \tfrac{64448}{6561}k_3 - \tfrac{212}{729}k_4\big)\big) \\
k_6 &= f\big(t_n + h,\ y_n + h\big(\tfrac{9017}{3168}k_1 - \tfrac{355}{33}k_2 + \tfrac{46732}{5247}k_3 + \tfrac{49}{176}k_4 - \tfrac{5103}{18656}k_5\big)\big) \\
k_7 &= f\big(t_n + h,\ y_n + h\big(\tfrac{35}{384}k_1 + \tfrac{500}{1113}k_3 + \tfrac{125}{192}k_4 - \tfrac{2187}{6784}k_5 + \tfrac{11}{84}k_6\big)\big)
\end{aligned}
$$

Fifth-order step (the one that is kept):

$$y_{n+1} = y_n + h\Big(\tfrac{35}{384}k_1 + \tfrac{500}{1113}k_3 + \tfrac{125}{192}k_4 - \tfrac{2187}{6784}k_5 + \tfrac{11}{84}k_6\Big)$$

Fourth-order check:

$$\hat{y}_{n+1} = y_n + h\Big(\tfrac{5179}{57600}k_1 + \tfrac{7571}{16695}k_3 + \tfrac{393}{640}k_4 - \tfrac{92097}{339200}k_5 + \tfrac{187}{2100}k_6 + \tfrac{1}{40}k_7\Big)$$

Error estimate used to pick the next step size:

$$\text{err} = \|y_{n+1} - \hat{y}_{n+1}\| = O(h^5), \qquad h_{\text{new}} = h\left(\frac{\text{tol}}{\text{err}}\right)^{1/5}$$

Note k_7 uses the fifth-order result, so it equals k_1 of the next step ("first same as last").

Butcher tableau

0							
$\frac{1}{5}$	$\frac{1}{5}$						
$\frac{3}{10}$	$\frac{3}{40}$	$\frac{9}{40}$					
$\frac{4}{5}$	$\frac{44}{45}$	$-\frac{56}{15}$	$\frac{32}{9}$				
$\frac{8}{9}$	$\frac{19372}{6561}$	$-\frac{25360}{2187}$	$\frac{64448}{6561}$	$-\frac{212}{729}$			
1	$\frac{9017}{3168}$	$-\frac{355}{33}$	$\frac{46732}{5247}$	$\frac{49}{176}$	$-\frac{5103}{18656}$		
1	$\frac{35}{384}$	0	$\frac{500}{1113}$	$\frac{125}{192}$	$-\frac{2187}{6784}$	$\frac{11}{84}$	
b (5th)	$\frac{35}{384}$	0	$\frac{500}{1113}$	$\frac{125}{192}$	$-\frac{2187}{6784}$	$\frac{11}{84}$	0
$\hat{b}$ (4th)	$\frac{5179}{57600}$	0	$\frac{7571}{16695}$	$\frac{393}{640}$	$-\frac{92097}{339200}$	$\frac{187}{2100}$	$\frac{1}{40}$